7/10/2026

Assignment-7

Created By
~pintu mer

1. **List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.**

IP Class	IP Address Range	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Large company network
Class B	128.0.0.0 – 191.255.255.255	University network
Class C	192.0.0.0 – 223.255.255.255	Small office or home network
Class D	224.0.0.0 – 239.255.255.255	Multicast video streaming
Class E	240.0.0.0 – 255.255.255.255	Experimental networking and research

Note: **Class A, B, and C are used for traditional unicast addressing. Class D is used for multicast, while Class E is reserved for experimental purposes. The addresses 127.x.x.x are reserved for loopback.**

2. **Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.Show your working steps for each calculation.**

Given Information

- **IP Address: 192.168.10.45**
- **Subnet Mask: 255.255.255.0**

Step 1: Network Address

The last number becomes 0.

Network Address: 192.168.10.0

Step 2: Broadcast Address

The last number becomes 255.

Broadcast Address: 192.168.10.255

Step 3: Valid Host Range

The first usable IP is 1, and the last usable IP is 254.

Valid Host Range: 192.168.10.1 – 192.168.10.254

Final Answer

Item	Answer
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
Valid Host Range	192.168.10.1 – 192.168.10.254
Total IP Addresses	256
Usable IP Addresses	254

3. **Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.**

Subnetting Plan for a Food Delivery Startup

We have the network 10.0.0.0/24 and need 4 separate subnets for Delivery Partners, Restaurants, Customers, and Admins.

Step 1: Divide the Network

- **Original network: 10.0.0.0/24**
- **Required subnets: 4**
- **Borrow 2 host bits because \(2^2 = 4\).**
- **New subnet mask: 255.255.255.192 (/26)**
- **Each subnet has 64 total IP addresses and 62 usable host addresses.**

Step 2: Subnetting Table

Group	Network Address	Subnet Mask	Valid Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Step 3: Conclusion

Each group has a separate subnet. This helps organize the network, manage IP addresses, and apply security rules between groups.

Remember: /26 provides 64 total IP addresses, of which 62 are usable for devices in each subnet.

4. **You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.Constraint: Show your calculation steps and final answers.**

Given Information

- **Network: 172.16.0.0/16**
- **Required subnets: At least 20**

Step 1: Calculate Borrowed Bits

We need at least 20 subnets.

- **2^4 = 16 — not enough.**
- **2^5 = 32— enough.**

So, we borrow 5 bits.

Step 2: Calculate the New Subnet Mask

Original prefix = /16

New prefix = 16 + 5 = /21

New Subnet Mask: 255.255.248.0

Step 3: Calculate Hosts per Subnet

Total bits in IPv4 = 32

Host bits = 32 - 21 = 11

Total IP addresses:

2^11= 2048

Usable hosts:

2048 - 2 = 2046

Final Answers

Item	Answer
Original Network	172.16.0.0/16
Required Subnets	20
Bits Borrowed	5
Total Subnets Created	32
New Subnet Mask	255.255.248.0
CIDR Prefix	/21
Usable Hosts per Subnet	2046

Conclusion: The college can create 32 subnets, with 2,046 usable host addresses per subnet, which meets the requirement of at least 20 subnets.

5. **A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.**
   ➔

The IP address 192.168.1.300 is invalid because each octet must be between 0 and 255. The number 300 is too large.

A valid IP address in this subnet could be 192.168.1.100, assuming it is not already in use and is within the network's valid host range.

Example: 192.168.1.100 is a valid IP address.